



Literature Map

⚙️ Status	Not started
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📌 Priority	Medium

Below is a **literature map of your thesis field updated to ~2024–2025**.

Think of it as the **"conceptual landscape"** where your thesis sits. This is exactly the type of structure that helps in:

- literature review chapters
- thesis defense slides
- positioning your contribution

I structured it so you can clearly show **where your thesis contributes**.

Literature Map – Wine Production Prediction and Climate Analysis

1. Climate and Viticulture Research

This is the **domain foundation**.

Research in this area studies how climate influences grape development, wine quality, and production variability.

Typical approaches:

- climate indices
- phenology models
- agronomic analysis
- regional climate projections

Key references:

- Climate Change and Global Wine Quality – Ashenfelter
- The impact of climate change on the winegrape vineyards of the Portuguese Douro region – Fraga

These studies demonstrate that:

- temperature
- precipitation
- solar radiation

strongly influence wine production and grapevine phenology.

This literature motivates the **problem itself**.

2. Statistical Yield Prediction Models

These are the **earliest quantitative models**.

Typical methods:

- linear regression
- generalized linear models
- climate indices

Goal:

predict wine production using aggregated climate indicators.

Strengths:

- interpretable
- grounded in agronomy

Limitations:

- limited ability to capture nonlinear relationships.
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3. Machine Learning Yield Prediction

This is the **modern predictive modeling stream**.

Methods used:

- Random Forest
- Support Vector Machines
- Gradient Boosting
- Neural Networks

Examples:

- Machine Learning Approaches for Grapevine Yield Prediction – Santos
- Climatic Drivers of Wine Production Identified Using Random Forest Models – Lopes

Advantages:

- strong predictive performance
- ability to capture nonlinear interactions

Limitations:

- often **black-box models**
- difficult to interpret for domain experts.

4. Explainable AI and Interpretable ML

This research direction emerged as a response to black-box models.

Goal:

develop models that produce **transparent explanations**.

Approaches include:

- feature importance
- SHAP values
- rule-based models
- interpretable decision trees

Reference:

- Interpretable Machine Learning – Molnar

Key idea:

interpretability is essential in **scientific domains**.

5. Rule-Based Data Mining

This area focuses on discovering **human-readable patterns**.

Origins:

association rule mining.

Foundational work:

- Mining Association Rules Between Sets of Items in Large Databases – Agrawal

These methods generate rules such as:

```
IF temperature > X AND rainfall < Y  
THEN production tends to increase
```

Advantages:

- interpretable
- actionable
- descriptive.

6. Subgroup Discovery

Subgroup discovery extends rule mining to identify **statistically interesting subsets**.

Instead of predicting outcomes, it identifies:

subpopulations where the target behaves differently.

Key papers:

- Subgroup Discovery with CN2-SD – Lavrač
- APRIORI-SD: Adapting Association Rule Learning to Subgroup Discovery – Klösgen

Recent extensions:

- Exceptional Model Mining – Duivesteijn
- Robust Subgroup Discovery – Lemmerich

Focus:

- interpretable knowledge discovery.
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7. Distribution Rules

This is the **specific methodological area of your thesis**.

Distribution rules extend association rules to **numeric targets**.

Key work:

- Discovering Associations with Numeric Variables – Jorge
- Pruning and Summarizing the Discovered Associations – Jorge

Advantages:

- no discretization required
 - exploration of **target distributions**
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8. Temporal Pattern Discovery

Recent work explores subgroup discovery in time-series data.

Examples:

- Temporal Exceptional Model Mining – Duivesteijn

This area is still developing.

Most work focuses on:

- sequential data
- temporal anomalies

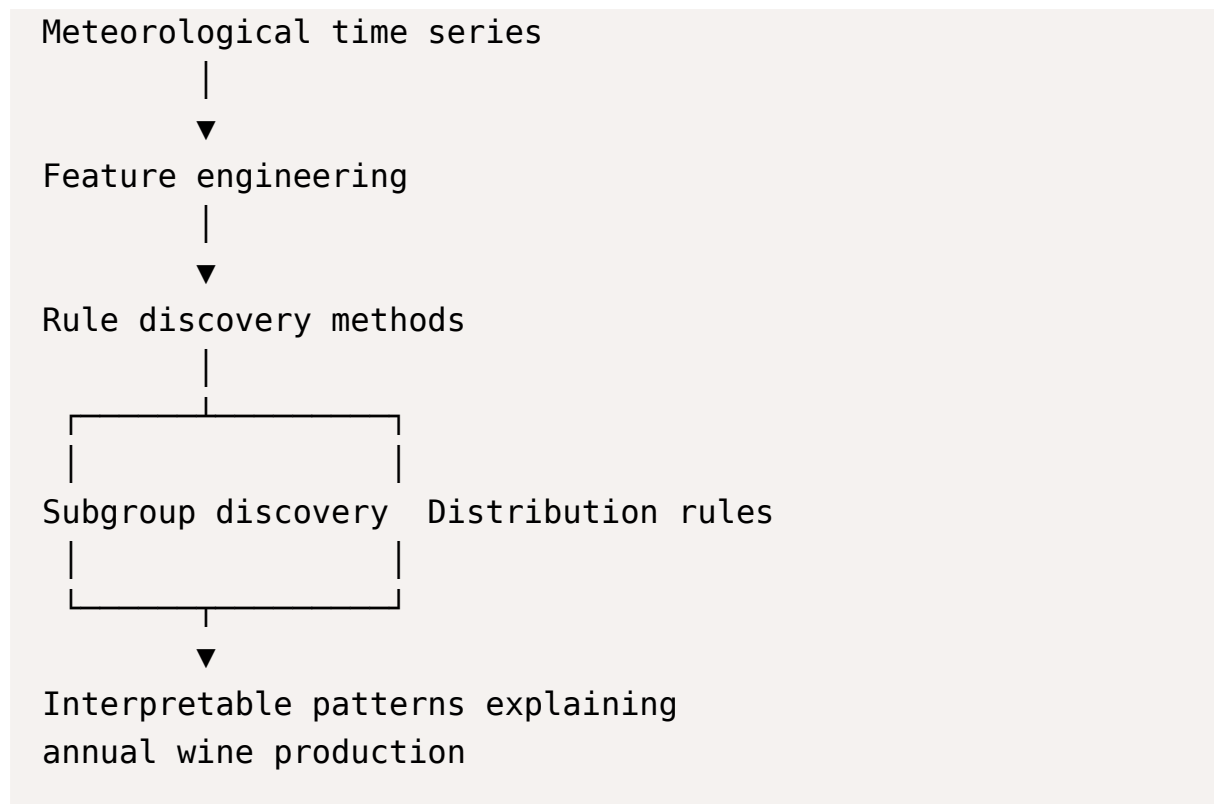
but not specifically on **meteorological feature engineering for annual targets**.

Where Your Thesis Fits

Your thesis connects **three research streams that rarely intersect**.

Climate science

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Why this positioning is strong

Your thesis contributes by combining:

- 1 **climate-based viticulture analysis**
- 2 **interpretable rule discovery**
- 3 **feature engineering from time-series meteorology**

Most existing work only covers **one or two of these areas**.

Final Research Gap (modern wording)

A good modern formulation is:

While machine learning approaches have been widely used to predict grape yield and wine production, relatively limited work has explored interpretable rule discovery techniques that transform long meteorological time series into human-readable patterns explaining annual wine production at the regional level.

If you want, I can also show you something very interesting

I can generate a **citation network of the subgroup discovery field**, showing:

- the **most cited papers**
- the **most recent work**
- the **research clusters**

This would help you ensure **your literature review is academically bulletproof**.